

When the Evaluator Decides the Result: Measuring Faithfulness in Retrieval-Augmented Generation

Shakhriyorbek Boltabaev Institute of Informatics, University of Szeged Szeged, Hungary shakhriyorbekboltabaev@gmail.com

Abstract

Faithfulness in retrieval-augmented generation is commonly measured by asking a natural language inference model whether the generated answer is entailed by the retrieved context, aggregated by taking the maximum over retrieved chunks. We hold a generation grid fixed — four embedding models, two question answering datasets, two generators, 16,000 generated answers — and vary only the faithfulness evaluator. Across four dataset-generator cells, the conclusion about whether embedding choice affects faithfulness changes in one of them depending on which evaluator is used. We trace this to measured properties of the evaluators rather than to the retrieval systems. Replacing a grounded value in an answer with one absent from the context — the failure a document question answering system most needs to detect — moves the entailment score by 0.054 on Natural Questions and 0.085 on HotpotQA for one generator, against 0.451 and 0.391 for the other, and is caught by a fixed threshold 5-14% of the time against 57-71%. The gap is largely accounted for by answer structure: the score falls as the number of assertions in an answer grows, and the two generators average 3.08 and 1.25 assertions per answer. It is not accounted for entirely — at a matched single assertion the two still differ, +0.102 against +0.485 — so answer structure is a partial account rather than a complete explanation. Claim-level aggregation and an alternative evaluator each recover part of the sensitivity but neither removes the effect. We also show that including refusals in the evaluation population produces an apparent relationship between retrieval quality and faithfulness that disappears when they are excluded. We report what these measurement choices do to a concrete equivalence claim, and give the conditions under which such a claim can be stated.

1 Introduction

A document question answering system is given a set of uploaded files and must answer from them. If an invoice records a project budget of \$1,000, an answer stating \$1,500 is a failure of a specific kind: it is fluent, on topic, cites the right document, and is wrong in the one place that matters. Systems of this kind are typically guarded by a faithfulness score — a measure of whether the generated answer is supported by the retrieved text — with answers below a threshold suppressed or flagged.

This paper asks what such a score actually measures. We do not propose a new evaluator. We take a fixed set of 16,000 generated answers and vary the measurement, and report where the measurement rather than the system determines the conclusion.

Our contributions are:

- A controlled falsification experiment. Replacing one grounded value in an answer with a value absent from the retrieved context, holding everything else fixed, gives a direct measurement of evaluator sensitivity to the error type that matters (Section 5).
- Evidence that the sensitivity depends strongly on answer structure, not only on the evaluator: entailment scores fall as the number of assertions per answer grows, for both generators tested (Section 5.3).
- A demonstration that the choice of evaluation population — specifically, whether refusals are included — can produce an apparent relationship between retrieval quality and faithfulness where none survives their exclusion (Section 6).
- A statement of what an equivalence claim about faithfulness requires: naming the evaluator, reporting the margin as a curve rather

than a point, and anchoring that margin to a measured effect size (Section 8).

The embedding-model grid is a testbed, not the object of study. It supplies a set of retrieval systems spanning 4.4 and 11.9 NDCG@5 points on the two datasets, with generation and prompting held constant, which is what makes the measurement comparison controlled.

2 Related Work

2.1 Retrieval quality and downstream outcome

Salemi and Zamani (2024) report that human document-level relevance labels, and LLM relevance judgments, correlate only weakly with downstream retrieval-augmented generation performance, and propose eRAG, which labels each retrieved document by the downstream result the system’s own generator produces from it alone. Their strongest baseline is answer-containment labelling, the scheme our qrels use, which retains a moderate correlation (Kendall’s tau 0.31-0.40 on the question-answering datasets); their reported correlations also decline as the number of retrieved documents grows, and are measured at $k = 50$ against ours at $k = 5$. Their downstream measure is correctness throughout, obtained from a fine-tuned 60M-parameter encoder-decoder reader. Our starting point is downstream of theirs: we take as given that retrieval metrics do not directly predict generation quality, and ask which downstream property a given measurement instrument can resolve at all — for faithfulness rather than correctness, and for prompted instruction-following generators, which can abstain.

2.2 Faithfulness measurement

Entailment-based faithfulness scoring is standard: a natural language inference model is asked whether the retrieved context entails the generated answer. RAGAS (Es et al., 2024) packages this alongside context and answer relevance metrics. AlignScore (Zha et al., 2023) trains a unified alignment function over several factual-consistency tasks rather than relying on a general-purpose inference model. FaithJudge (Tamber et al., 2025) uses a language model as the judge. Our work is orthogonal to proposing another such metric: we measure how three of them respond to a controlled, known error, and report the consequences for conclusions drawn with them.

2.3 Meta-evaluation of faithfulness detectors

A separate line of work evaluates the detectors themselves. TRUE (Honovich et al., 2022) standardises factual consistency evaluation over eleven annotated datasets and finds inference-based and question-generation approaches to be strong and complementary. AGGREFACT (Tang et al., 2023) aggregates nine summarisation factuality datasets, stratifies them by the model that produced the errors, and reports that no single metric is best in all settings or for all error types. RAGTruth (Niu et al., 2024) supplies word-level hallucination annotations over roughly 18,000 retrieval-augmented responses. FaithBench (Bao et al., 2025) is assembled from summaries on which current detectors disagree, and reports detection accuracies near 50% on them. MiniCheck (Tang et al., 2024) unifies recent grounding datasets as LLM-AGGREFACT and trains compact fact-checkers against that benchmark.

These works compare detectors against gold hallucination labels on a labelled corpus, and the quantity they report is detector accuracy. Our design contains no gold hallucination label. We hold the generation grid, the queries and the generated answers fixed, swap the evaluator, and ask whether the ranking of the systems under study survives. The two questions come apart: a detector that is more accurate on a labelled corpus need not be one that resolves differences between the systems a practitioner is actually choosing between, and it is the second property that decides whether a published comparison describes the systems or the instrument. Xiao et al. (2023) argue on general grounds that generation metrics should be analysed as measurement instruments with stated reliability and validity. The falsification probe in Section 5 and the equivalence curve in Section 8 are two such analyses, carried out on the instruments a RAG faithfulness result is usually reported with.

2.4 Hallucination and grounding analysis

ReDeEP (Sun et al., 2025) localises hallucination to attention heads and feed-forward components. Sinha (2025) reports that embedding-based detection separates synthetic hallucinations well and real ones poorly, the hard cases being those that remain semantically close to a faithful response. Chen et al. (2025) vary the embedding model in a RAG pipeline and combine several of them for mathematics question answering, on the observa-

tion that different embedders succeed on different queries. Our analysis is correlational and behavioural rather than mechanistic: we do not probe model internals.

3 Definitions

3.1 Faithfulness

For a generated answer a and retrieved context $R = \{d_1, \dots, d_k\}$, the standard aggregate is:

$$F_{\max}(a, R) = \max_{d \in R} \text{NLI}(d \models a) \quad (1)$$

We also evaluate a claim-level aggregate. Writing $C(a)$ for the assertions in a :

$$F_{\text{claim}}(a, R) = \min_{c \in C(a)} \max_{d \in R} \text{NLI}(d \models c) \quad (2)$$

The maximum over chunks is appropriate — any retrieved document may support a claim. The maximum over claims, implicit in (1), is not: an answer is only as supported as its least supported assertion, so one unsupported claim should not be masked by several supported ones. Faithfulness is measured against the generated answer and the retrieved context only; the gold answer is never used.

3.2 Correctness and retrieval hit

Correctness is measured against the gold answer and never against the context, so that a grounded wrong answer and an ungrounded right answer remain distinguishable. A retrieval hit means the gold answer string appears in the retrieved context. This is a strong definition of retrieval success, and the Limitations section notes what it does not capture.

3.3 On differencing retrieval quality and faithfulness

An earlier version of this work defined a gap metric as the difference between NDCG@5 and faithfulness, with a normalised variant dividing by retrieval quality. We report the properties of that family here because they are general rather than specific to our implementation. First, the difference has no common unit: a rank-based retrieval metric and an entailment probability are not commensurable. Second, on our data the difference is negative for every model on Natural Questions, since faithfulness (0.85) exceeds retrieval quality

Embedding model	Paradigm	NDCG@5 (NQ)	NDCG@5 (HotpotQA)
all-mpnet-base-v2	contrastive	0.786	0.705
BGE-M3	multilingual	0.794	0.809
E5-large-instruct	instruction-tuned	0.806	0.824
text-embedding-3-small	proprietary	0.830	0.767
spread		4.4 pts	11.9 pts

Table 1: Retrieval quality of the four systems in the testbed. The ranking differs between datasets, and the spread is far larger on HotpotQA.

(0.79) — the metric presumes generation loses fidelity relative to retrieval, and it does not. Third, any metric of the form $1 - F/RQ$ reproduces the retrieval ranking whenever F is near-constant across systems, so apparent agreement of such a metric across generators is uninformative: retrieval quality is generator-invariant by construction. We therefore report absolute faithfulness throughout. Retiring the metric also disposes of a known objection to it at the root rather than patching it: a difference cannot separate a system whose retrieval and faithfulness are both high from one where both are low, and the normalised variant rescales that ambiguity without removing it. Reporting the two quantities separately keeps the both-low case visible. The retired implementation is held out of the analysis path that produced the tables reported here.

4 Experimental Setup

Two question answering datasets are used, Natural Questions (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018), at 1,000 queries each. Four embedding models index each corpus; retrieval takes the top five chunks of 256 tokens with 32-token overlap; two generators produce an answer from a byte-identical prompt at temperature 0. This yields $4 \times 2 \times 2 \times 1,000 = 16,000$ generated answers, each re-scored by every evaluator. Prompt parity across generators is enforced by construction — a single prompt builder, one user turn, no system message — because a prompt difference would confound every cross-generator comparison in Section 5.

Three evaluators are compared. NLI-max is (1) with a DeBERTa-v3-large (He et al., 2023) inference model, the common choice. Claim-min is (2) with the same inference model and sentence-level claim segmentation, chosen so that no second generator enters the measurement loop. AlignScore (Zha et al., 2023) is an independently trained alignment model. All three score identical inputs.

The answers are byte-identical across evaluators; the premises are not, and the difference is

part of what the comparison measures. Premise construction follows each evaluator’s intended interface. The entailment scorer receives each of the five retrieved chunks separately plus their concatenation truncated to the model’s 512-token window, and the maximum is taken, since any one retrieved document may entail the answer and the concatenation exceeds the window. The claim-level scorer applies that same construction to each extracted claim. AlignScore receives the concatenated context as a single premise and performs its own splitting internally. Part of the difference reported in Section 5 is therefore premise granularity rather than evaluator sensitivity; Section 5.1 bounds that share by re-scoring the same cases with the same entailment model on the concatenated premise alone. The hypothesis is normalised identically for all three: markdown emphasis, headings and list markers are stripped before scoring. This matters because one generator emits markdown in 74.9% of its answers and the other in 0.3%, and the entailment model responds to it — see Section 7.2.

One refusal rule is used throughout. An answer counts as a refusal when, after SQuAD-style normalisation, it begins with one of a fixed list of declining phrases; a single shared implementation applies it, so Sections V through VIII describe the same population. It excludes 4,123 of the 16,000 answers (25.8%): 587 on NQ/Claude (14.7%), 1,162 on NQ/GPT-4o-mini (29.0%), 1,166 on HotpotQA/Claude (29.1%) and 1,208 on HotpotQA/GPT-4o-mini (30.2%). Both generators prepend a "Based on the provided context," preamble to everything, refusals included, so the rule strips that preamble before the anchored test; without it 107 refusals, all of them Claude’s, would be graded as attempts.

5 How Sensitive Are These Evaluators to a Known Error?

5.1 A controlled falsification

A faithfulness score is only useful if it responds when an answer stops being faithful. We construct that condition directly. For each answer containing a numeric value that appears in the retrieved context, we replace that value with one that does not appear in the context, preserving surface format, and re-score against the same chunks. Two eligibility rules are load-bearing: the original value must be present in the context, or the answer was

never grounded in it; and the replacement must be absent, or the altered answer is accidentally supported and a low response is correct behaviour.

Cases are taken in dataset order rather than drawn at random: for each system, dataset and generator we use the first 200 eligible answers. The scored set is therefore a prefix of an already filtered population, and 3,184 cases are scored against 5,272 eligible across the grid — 800 of 1,946 on NQ/Claude, 800 of 882 on NQ/GPT-4o-mini, 800 of 1,551 on HotpotQA/Claude and 784 of 893 on HotpotQA/GPT-4o-mini. The two GPT-4o-mini cells are close to a census; the two Claude cells are sampled at two fifths and one half.

Two controls accompany it. An entity substitution replaces a grounded named entity instead of a value, giving a comparable edit. Scoring the untouched answer against another query’s retrieved context gives the floor of each evaluator’s scale on this data. 200 paired cases per cell, rebuilt deterministically so all three evaluators see identical inputs.

On Claude’s Natural Questions answers, NLI-max leaves the falsified answer at 0.852 against a floor of 0.474: the altered answer remains comfortably above any usable threshold. On GPT-4o-mini’s answers the same edit lands at 0.436, essentially at that generator’s floor of 0.506. The evaluator, the context and the perturbation code are identical; only the generator that wrote the answer differs.

Detection rates are only comparable across evaluators after the scale is normalised, because the evaluators do not share a floor, and we therefore report two gates. On this cell an untouched answer scored against an unrelated context averages 0.474 under NLI-max against 0.245 under claim-min, so a fixed 0.5 gate sits just above the floor for one evaluator and well above it for the other. The random-context distribution is also heavy-tailed — its 95th percentile is 0.985 under NLI-max — so an unrelated document clears 0.5 nearly half the time. The second gate is anchored at that 95th percentile, at which "detected" means the falsified answer is no longer better supported than an unrelated document, which is the same statement on every scale. Read that way NLI-max detects 50.2% of falsifications on Claude’s NQ answers rather than 5.1%, and its gap to GPT-4o-mini narrows from roughly elevenfold to under twofold. The ordering of the three evaluators is unchanged at either gate. The floor-anchored gate is a compari-

Dataset / generator	Evaluator	Orig	Falsified	Floor	Detected @0.5	Detected @floor gate
NQ / Claude Haiku 4.5	NLI-max	0.906	0.852	0.474	5.1% [3.8-7.0]	50.2% [45.7-54.7]
	Claim-min	0.679	0.514	0.245	25.1% [21.7-28.9]	58.3% [53.3-63.1]
	AlignScore	0.866	0.619	0.331	24.1% [21.3-27.3]	77.9% [74.4-81.1]
NQ / GPT-4o-mini	NLI-max	0.887	0.436	0.506	56.6% [52.9-60.1]	85.2% [81.9-88.1]
	Claim-min	0.863	0.390	0.490	61.0% [57.4-64.6]	91.0% [88.1-93.2]
	AlignScore	0.912	0.272	0.160	75.4% [72.2-78.4]	82.1% [79.2-84.7]
HotpotQA / Claude	NLI-max	0.702	0.618	0.427	13.9% [11.3-17.0]	49.7% [42.3-57.2]
	Claim-min	0.370	0.255	0.161	34.3% [29.0-40.0]	48.9% [41.8-56.1]
	AlignScore	0.786	0.454	0.475	48.9% [45.2-52.5]	93.7% [90.7-95.8]
HotpotQA / GPT-4o-mini	NLI-max	0.577	0.186	0.201	70.5% [66.2-74.6]	83.8% [79.7-87.2]
	Claim-min	0.571	0.182	0.204	70.6% [66.2-74.6]	84.2% [80.1-87.6]
	AlignScore	0.780	0.119	0.234	89.4% [86.8-91.6]	97.5% [95.8-98.5]

Table 2: Response of three evaluators to a falsified value, on identical inputs, pooled over the four embedding systems ($n = 800$ per cell, 784 on HotpotQA/GPT-4o-mini). "Detected" is the share of answers that passed a threshold before falsification and fail it after, with Wilson 95% intervals. "Floor" is the mean score of an untouched answer against an unrelated query's context; the floor gate is the 95th percentile of that same distribution, per cell and evaluator.

son device and not an operating point: it sits above the mean untouched answer, so a deployed system could not use it.

5.2 The effect is not verbatim copying

A natural explanation is that one generator quotes the retrieved text back, so the evaluator scores a copy of its own premise and a single substituted value cannot move it. The data does not support this. Over the 3,173 falsification cases, the fraction of answer word 5-grams occurring in the retrieved context as a contiguous token sequence is 0.226 for Claude and 0.235 for GPT-4o-mini: the generator whose scores barely move copies slightly less, not more. The correlation between overlap and the response to falsification is +0.046 for Claude (95% CI -0.004 to +0.096) and +0.166 for GPT-4o-mini (+0.118 to +0.212) — indistinguishable from zero in the first case and positive in the second, where the explanation requires both to be negative.

The comparison also survives at matched overlap, which the correlation alone does not show. Grouping the cases into four overlap bands leaves Claude between +0.052 and +0.085 and GPT-4o-mini between +0.326 and +0.507; the separation is at least fivefold in every band, including the band above 0.3 where both generators are reusing a substantial part of the retrieved wording. Overlap is measured on tokens with punctuation mapped to spaces, so it is unaffected by the markdown normalisation described in Section 7.2.

5.3 Answers with more assertions are scored less sensitively

Segmenting each answer into sentence-level assertions and grouping by count gives a declining relationship, present for both generators. It is not strictly monotone: the tail buckets are thin and both generators reverse direction once within them.

The two generators average 3.08 and 1.25 assertions per answer, and median answer lengths of 55 and 15 words. This largely accounts for the cross-generator difference in Table II: a falsified value is a progressively smaller edit to the hypothesis as correct surrounding text accumulates, and the entailment judgement is dominated by the remainder. It does not account for it entirely — within the single-assertion bucket the two generators still differ, +0.102 against +0.485. As a check on the claim-level implementation, every single-assertion answer scores identically under both aggregates ($n = 1,640$, largest absolute difference $1.7\text{e-}05$, which is batch-padding noise), the expected degenerate case. That check is what surfaced the normalisation defect reported in Section 7.2: before markdown was stripped for both aggregates it failed on every one of the 131 formatted answers, by as much as 0.729.

Claim-level aggregation is a partial remedy rather than a repair. Expressed as a fraction of the distance to the random-context floor, the median response on Claude's answers moves from 0.01 to 0.03, and the minimum moves in the correct direction in 55% of cases: when the falsified assertion is not already the weakest, the minimum does

Assertions	n (Claude)	Claude NLI-max	Claude Claim-min	n (GPT)	GPT NLI-max	GPT Claim-min
1	91	+0.102	+0.102	680	+0.485	+0.485
2	300	+0.076	+0.151	77	+0.360	+0.467
3	194	+0.032	+0.229	26	+0.041	+0.401
4	103	+0.024	+0.162	10	+0.204	+0.192
5 or more	112	+0.024	+0.146	7	+0.032	+0.002

Table 3: Drop in score when one grounded value is falsified, by number of assertions in the answer, on Natural Questions. Under NLI-max the response decays as answers accumulate assertions; under claim-level aggregation it does not. The GPT-4o-mini rows beyond two assertions rest on 26, 10 and 7 cases and should be read as such. In the single-assertion row the two aggregates coincide exactly, which is the degenerate case they must reduce to.

HotpotQA, Claude	all-mpnet	text-emb-3-small	BGE-M3	E5-instruct
NDCG@5	0.705	0.767	0.809	0.824
Refusal rate	0.376	0.279	0.224	0.207
Faithfulness, all rows	0.548	0.599	0.628	0.638
Faithfulness, answered	0.737	0.727	0.731	0.746

Table 4: Pooling refusals into the faithfulness average reproduces the retrieval ranking exactly (spread 0.090). Restricted to answered rows the spread is 0.019 and the ordering does not follow retrieval quality.

not move at all. A second dilution remains within assertions — at matched assertion count the two generators still differ (+0.102 against +0.485), and their assertions average 20.5 against 13.8 words.

6 The Evaluation Population Can Manufacture a Relationship

Both generators decline to answer on a substantial share of queries, and the rate depends on retrieval quality. A refusal is correctly not entailed by the context and scores near the floor. Including refusals therefore transfers a retrieval effect into the faithfulness average, at no cost in plausibility.

Any evaluation population whose membership depends on the system under comparison can transfer an effect from the selection into the measurement. Here the weakest retriever refuses most often, and every refusal lowers its average. All faithfulness results in this paper exclude refusals and are paired on the queries every system answered.

6.1 Membership Is Set by a Grader, and Graders Disagree

Excluding refusals requires deciding which answers are refusals, and that decision is made by a rule rather than given by the data. We compare two defensible rules: a string-matching heuristic keyed to the refusal templates the prompts offer, and a judge model shown the question, the reference answers and the answer. They agree on the

HotpotQA, Claude	all-mpnet	text-emb-3-small	BGE-M3	E5-instruct
Refusal rate, heuristic	0.376	0.279	0.224	0.207
Refusal rate, judge	0.422	0.332	0.277	0.252

Table 5: Two refusal detectors on the same answers. The judge finds roughly four to five more refusals per hundred answers in every column, so the disagreement is a level shift rather than noise, and it does not reorder the systems.

large majority of rows and disagree consistently at the margin.

A uniform shift that preserves the ordering would ordinarily be unremarkable. It is not unremarkable here, because the excluded rows are not a random sample of the population: they are the rows nearest the boundary between refusing and attempting. Substituting one detector for the other moves nineteen rows on Natural Questions with Claude and fifty-five on HotpotQA with Claude, and changes the significance verdict in two of the eight dataset-generator-evaluator cells. Under the entailment aggregate, Natural Questions with Claude moves from no detectable difference ($p = 0.223$) to a detectable one ($p = 0.048$); under AlignScore, HotpotQA with Claude moves from $p = 0.151$ to $p = 0.049$. Both land within a thousandth of the conventional threshold, which is the point: the verdict in these cells is not a property of the systems.

The count is more stable than the cells. One of four dataset-generator cells changes verdict between evaluators under either detector, but not the same two. We therefore report the count as the finding and treat any individual cell as undetermined.

6.2 The Correctness Grader Admits Refusals to the Wrong Population

The same problem appears once faithfulness is conditioned on correctness. String containment grades an answer correct when a reference answer

string occurs in it, and a short reference string can occur inside a refusal. On a 200-answer calibration, ten refusals were graded correct by containment and the judge overturned nine of them. A refusal admitted to the correct population carries a near-floor faithfulness score into the correct-answer average, which lowers the baseline that any correct-versus-incorrect comparison is measured against. Under graded judgement no refusal is graded correct in any of the sixteen cells.

The effect is large enough to change a sign. Measured with containment and a baseline that excluded refusals from the incorrect side only, answers on Natural Questions with Claude appeared more grounded when wrong than when right, by 0.018 to 0.050 across the four systems. Excluding refusals from both sides reduces this to between -0.006 and +0.018. Grading correctness with the judge instead of containment moves it to between -0.051 and -0.113, in the opposite direction. Each of the three steps is a correction to the grader rather than a change to the answers, and all three move the quantity the same way.

Tested directly, no positive value of this quantity survives. Across all sixteen cells a two-sample permutation test with Holm correction finds three significant, all negative, and none of the eight positive-signed cells has a confidence interval excluding zero. The clearest single result is E5-instruct on Natural Questions, where the same effect appears under both generators (-0.113 and -0.115). We report no evidence that answers are more grounded when they are wrong. The intervals are wide, between 0.08 and 0.13, because only 44 to 84 wrong answered rows remain per cell once refusals are excluded; these tests bear on the general claim and not on individual systems.

6.3 A Contingency Cell That Mostly Counts Refusals

The same accounting governs the contingency between retrieval success and correctness. The cell in which retrieval succeeded and the answer was still wrong is the one that bears on whether retrieval quality is sufficient, and most of it consists of answers that were never attempted.

Read without the decomposition, the raw cell would support the claim that good retrieval frequently fails to produce a correct answer. Most of what it counts is a system declining to answer, and refusal rates differ by more than twenty points across the systems compared. The residual is a

Retrieval hit and answer incorrect	Claude, NQ	Claude, HQA	GPT, NQ	GPT, HQA
Share of all queries	17.4%	34.0%	31.7%	36.3%
Of that cell, refusals	69.3%	86.8%	79.0%	78.1%
Attempted and wrong	5.4%	4.5%	6.7%	7.9%

Table 6: Correctness from graded judgement. The raw cell is between 17 and 36 per cent of queries; the share that is a grounded, committed, wrong answer is between 4.5 and 7.9 per cent.

real phenomenon and it is roughly a fifth the size the raw cell implies.

7 The Verdict Depends on the Evaluator

We now compare the four retrieval systems on faithfulness, refusals excluded, paired on a common query subset, with a paired bootstrap of 10,000 resamples.

The statistic is the spread: the difference in mean faithfulness between the best and the worst of the four systems, with the p-value from a paired bootstrap over that pair, 10,000 resamples, re-centred at zero (Dror et al., 2018). The range is the quantity the decision this table informs is exposed to — a practitioner adopts one embedding model, and what they stand to lose is the distance to the best alternative, not the dispersion of the set. It is however a summary of two systems selected for being the extremes, so the p-value beside it is not adjusted for that within-cell selection, and the two intermediate systems do not appear in it at all. We therefore treat the range as descriptive, and report the direction across all four systems separately in Table VIII rather than reading it off the extremes. An omnibus test over the four systems would be the stricter statistic; we do not run one here.

Under the common choice of evaluator, no cell shows a detectable difference. Under an evaluator measured in Section 5 to be more responsive to known errors, one does, and a second is significant before correction and not after. A result reported only under the first evaluator would be an artifact of instrument resolution rather than a property of the systems. The effect is narrower than the four-cell design can establish, and we report it as one cell rather than as a general property of the grid.

That count of one is itself not invariant to the correction. NQ/Claude under AlignScore is at $p = 0.0081$ uncorrected, and Holm places it either side of $\alpha = 0.05$ depending on which comparisons are treated as one family: 0.057 over the eight in the table, 0.024 over the four sharing an evaluator, 0.016 over the two in its own cell. Correcting

Dataset / generator	n	NLI-max spread	p (Holm)	AlignScore spread	p (Holm)
NQ / Claude	784	0.0107	0.155 (0.391)	0.0114	0.0081 (0.0567)
NQ / GPT-4o-mini	628	0.0130	0.098 (0.391)	0.0080	0.117 (0.391)
HotpotQA / Claude	517	0.0232	0.159 (0.391)	0.0164	0.063 (0.380)
HotpotQA / GPT-4o-mini	510	0.0267	0.065 (0.380)	0.0461	0.0005 (0.0040)

Table 7: The same comparison under two evaluators, with the uncorrected p first and the Holm-adjusted p in parentheses, corrected over the eight comparisons in the table. One of four cells changes verdict at $\alpha = 0.05$ and survives correction; NQ/Claude is significant uncorrected and not after correction. Queries, answers, refusal handling and test are identical throughout. With 10,000 resamples the smallest attainable non-zero p is $1e-4$, so the two significant cells at $p = 0.0005$ are 5 extreme resamples of 10,000 and not a clamped floor.

Dataset	Claude / NLI	Claude / Align	GPT / NLI	GPT / Align
Natural Questions	-1.00	-0.40	-0.80	0.00
HotpotQA	+0.40	+0.80	+0.80	+1.00

Table 8: With four systems these coefficients cannot carry significance individually. The pattern is that all four HotpotQA cells are positive and all four Natural Questions cells are zero or negative, across both generators and both evaluators.

within evaluator is defensible, since the four NLI tests establish which cells are null and so precede the question rather than competing with it, and it would make the count two. We report the eight-comparison family because it is the most conservative of the three and because it was fixed before the correction was applied; adopting a smaller family after observing that it restores a cell would be a selection we could not account for. We state the alternative rather than only the choice, because a count that moves with the multiplicity family is the same kind of dependence on an analysis decision that this paper reports for the choice of evaluator.

7.1 Direction is dataset-dependent

Where a difference exists, it does not point the same way. Spearman correlation between NDCG@5 and faithfulness across the four systems, computed in every generator-evaluator combination:

On the multi-hop dataset, better retrieval accompanies more faithful answers; on the single-hop dataset it does not. A plausible reading is that composing an answer from several retrieved pieces makes grounding sensitive to how much of the needed evidence is present, while a single-fact answer is not. We do not test this, and note that the experiment varies embedding model rather than manipulating retrieval quality directly, so the correlation should not be read as an effect of NDCG@5 itself.

7.2 A normalisation defect that produced a result

An earlier version of this analysis reported two cells rather than one. The difference was not a change of test, data or population but of text normalisation, and it is worth reporting because it is an instance of the paper’s own subject.

The entailment scorer originally read the answer exactly as the generator wrote it, while the claim-level scorer stripped markdown from each claim before scoring. The two aggregates were therefore not reading the same hypothesis. The discrepancy is invisible in aggregate and was found by the degenerate-case check in Section 5.3: with a single claim and no markdown the two must be numerically identical, and they were, to $1.7e-05$; with markdown they differed on every one of 131 cases, by up to 0.729 on a pair of asterisks alone. Markdown is not evenly distributed across the grid — it appears in 74.9% of Claude’s answers and 0.3% of GPT-4o-mini’s — so the formatting habits of one generator were entering a comparison between embedding systems.

Stripping markdown for all three evaluators removes the asymmetry and restores the identity on every single-assertion answer. It also removes the significance of the NQ/Claude cell under AlignScore: its spread falls from 0.0179 to 0.0114 and its Holm-adjusted p from 0.0040 to 0.0567. That cell’s apparent evaluator-dependence was substantially an artifact of formatting, and the count reported above is the corrected one. We record it rather than silently reporting the smaller number because the failure mode generalises: a preprocessing choice applied to one aggregate and not another is invisible in every summary statistic, survives every significance test, and is detectable only by an identity that the two aggregates must satisfy. Metrics that admit such an identity should be checked against it.

8 What an Equivalence Claim Requires

Two of the four cells in Table VII show no detectable difference under either evaluator. Absence of a detected difference is not evidence of equivalence, so we test equivalence directly with two one-sided tests (Schuirmann, 1987). A TOST result depends entirely on the margin, and a margin is a substantive judgement rather than a statistical fact — so we report a curve rather than assert a point.

Section 5 supplies an external anchor for choosing that column. Falsifying one grounded value moves NLI-max by 0.049 to 0.095 on Claude’s answers. A margin of ± 0.05 on that evaluator is therefore approximately the size of one fabricated fact, which is not a negligible difference. Stating equivalence at ± 0.05 without that context asserts more than the instrument supports.

9 Discussion

For work that reports RAG faithfulness, we suggest the following.

- Name the evaluator in the claim. "Faithfulness did not differ" is incomplete; two of our four cells reverse between evaluators on identical data.
- Report the evaluation population explicitly, and whether membership depends on the system being compared. Pooling refusals reproduced the retrieval ranking exactly here (Table IV).
- Prefer claim-level aggregation to a maximum over the whole answer, and expect it to help most where answers carry several assertions.
- Calibrate the margin against a measured effect before claiming equivalence. A falsification experiment of the kind in Section 5 costs nothing beyond re-scoring.

For a document question answering system, the practical reading of Table II is that a grounding threshold on a verbose generator detects a minority of falsified values at a fixed gate: 5.1% under the common evaluator, 25.1% under the most responsive one tested. A deterministic check that every value in an answer appears in the retrieved context addresses this class of error directly and is complementary to an entailment score. Run over the same cases as Table II, it recalls every falsified value

in all four cells at a false-positive rate of 0.7% to 5.2% on untouched answers, against 5.1% detection for NLI-max on the hardest of those cells. Two qualifications matter. Numbers that are not assertions about the world — the chunk indices the prompt supplies and the markers of an ordered list — are neither asserted of the world nor expected in the retrieved text, and counting them raises the false-positive rate on Claude’s HotpotQA answers from 5.2% to about 39%. And the residual false positives are quantities the answer derives rather than copies: counts, sums, elapsed years, formats converted. A literal check flags all of those, so it is a complement to an entailment score and not a replacement for one.

Checking symbolic content against the source is not a new proposal, and we do not claim it as one. Goodrich et al. (2019) score generated text by extracting subject-relation-object triples and comparing them against the source document. Nan et al. (2021) define entity-level consistency metrics for summarisation and evaluate them beside entailment-based alternatives. Zhao et al. (2020) address quantity hallucination directly, verifying dates, numbers and monetary amounts in generated summaries against the source. That numbers are a weak point is also documented at the level of the representations these systems are built on: Wallace et al. (2019) find number magnitude encoded unevenly across standard embeddings and least reliably by subword models. Entailment is separately a loose proxy for attribution — the AIS framework (Rashkin et al., 2023) distinguishes whether a statement is supported by an identified source from whether it is plausible or entailed in a general sense, which is the distinction a structurally similar but differently attributed claim exploits.

What this paper adds is narrower than the check itself: the controlled minimal pair. A single numeric value is replaced in an otherwise byte-identical generation, and the same case is scored by all three evaluators, so the quantity measured is each evaluator’s response to a known error on identical input rather than its agreement with a human label on some other corpus. The scope of that measurement is numeric values, two proprietary generators and two datasets.

Dataset / generator	Evaluator	± 0.01	± 0.02	± 0.03	± 0.05	± 0.10
NQ / Claude	NLI-max	0/6	4/6	6/6	6/6	6/6
	AlignScore	2/6	3/6	6/6	6/6	6/6
NQ / GPT-4o-mini	NLI-max	0/6	3/6	6/6	6/6	6/6
	AlignScore	1/6	6/6	6/6	6/6	6/6
HotpotQA / Claude	NLI-max	0/6	0/6	0/6	6/6	6/6
	AlignScore	0/6	3/6	6/6	6/6	6/6
HotpotQA / GPT	NLI-max	0/6	0/6	1/6	4/6	6/6
	AlignScore	0/6	0/6	0/6	4/6	6/6

Table 9: System pairs judged equivalent, of six tested, across margins. A claim of equivalence is a claim about a particular column.

10 Conclusion

Holding a 16,000-answer generation grid fixed and varying only the faithfulness evaluator changes the conclusion about whether embedding choice affects faithfulness in one of four dataset-generator cells. The differences between evaluators are measurable directly: a falsified grounded value moves the common entailment aggregate by 0.054 on one generator and 0.451 on the other, and that gap is largely accounted for by how many assertions an answer contains rather than by the retrieval system. Reported without naming the evaluator, the evaluation population, the aggregation and the equivalence margin, a faithfulness result is not reproducible in the sense that matters — another group applying a different standard choice to the same generations would reach a different conclusion.

Limitations

The claim is bounded by the grid it was measured on and by the graders that define its populations.

- Four embedding models, two English datasets and two proprietary generators. The generator comparison in particular rests on two systems whose answer-length distributions differ substantially; an open-weight generator with a third profile would test the mechanism in Section 5.3 rather than merely repeat it.
- Correctness is graded by a judge model rather than by string overlap. Exact match is 0.000 across all 16,000 answers because the generators do not emit bare answer spans. String containment, the obvious fallback, is not a bound in either direction: against the judge it understates accuracy by 5 to 15 points in twelve cells and overstates it by 3 to

7 points in the four HotpotQA cells under Claude, where long answers mention a reference string while asserting something else. The judge is itself a language model applied to its own outputs, and its agreement with a human annotator on this data has not been established.

- Retrieval hit is defined as the gold answer string appearing in the retrieved context. This is a strong definition, and it does not separate retrieval failure from failures of composition, entity resolution or inference once the evidence is present.
- The Natural Questions sample retains only queries whose annotated answer falls inside the retained context window. This makes the retrieval task more constrained than open-domain RAG, where evidence may be distributed, partial or implicit.
- Pairing on the queries every system answered removes exactly those queries where retrieval quality determined whether an answer was produced. The reported comparison is therefore conditional on that subset, which is narrower than an unconditional comparison.
- Effect sizes are small where detected — 0.018 and 0.046 — against retrieval spreads of 4.4 and 11.9 NDCG@5 points. These are detectable differences, not large ones.
- The three evaluators receive byte-identical answers but not identical premises. The entailment scorer is given each of the five retrieved chunks separately, plus their concatenation truncated to the model’s window, and the maximum is taken; the claim-level scorer applies that construction per extracted claim; AlignScore is given the concatenated context

as one premise and performs its own splitting. Each is the evaluator used as its authors intend, but it means part of the difference reported in Section 5 is premise granularity rather than evaluator sensitivity. Rescoring the same cases with the same entailment model on the concatenated premise bounds that share between none and half of the gap depending on the cell, and at about a quarter of it on Claude’s answers. The hypothesis, by contrast, is now normalised identically for all three (Section 7.2); it was not in an earlier version, and that difference alone decided one cell.

- Which answers count as refusals is fixed by a rule rather than given by the data, and that rule sets the evaluation population for every result in Sections V through VIII. It is therefore a researcher degree of freedom as well as a finding: substituting one defensible detector for another moves the rows nearest the boundary between refusing and attempting, and carries the uncorrected p-value across the conventional threshold in two of eight cells (Section 6.1). Those two do not survive correction for multiple comparisons, but the population every reported number is computed on still depends on the choice. The rule used here strips the generators’ standard preamble before matching; without that step 107 refusals, all from one generator, would be graded as attempts.
- The falsification cases are the first 200 eligible answers in dataset order for each system, dataset and generator, rather than a random sample of the eligible answers: 3,184 cases scored against 5,272 eligible across the grid. Eligibility requires a grounded numeric value in the answer together with a replacement value absent from the context, so the scored set is a prefix of an already filtered population. Cells whose eligible count is near 200 are close to a census; the largest cells are sampled at about two fifths.
- The repository configures seven embedding models and this paper reports four: all-mpnet-base-v2 (contrastive), BGE-M3 (multilingual), E5-large-instruct (instruction-tuned) and text-embedding-3-small (proprietary), chosen to span pretraining paradigms

and to include one closed-source arm. GTE-large, Instructor-XL and jina-embeddings-v3 (Sturua et al., 2024) are configured and runnable but were not run at this scale: each additional model is a further generation pass over both datasets and both generators, and the grid is a testbed for the evaluator comparison rather than a survey of embedding models.

- The correct-versus-incorrect faithfulness comparison rests on 44 to 84 attempted wrong answers per cell. Excluding refusals and grading correctness with a judge both reduce that population, and the resulting confidence intervals are 0.08 to 0.13 wide. The tests reported in Section 6.2 bear on the general claim and do not resolve individual systems.

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